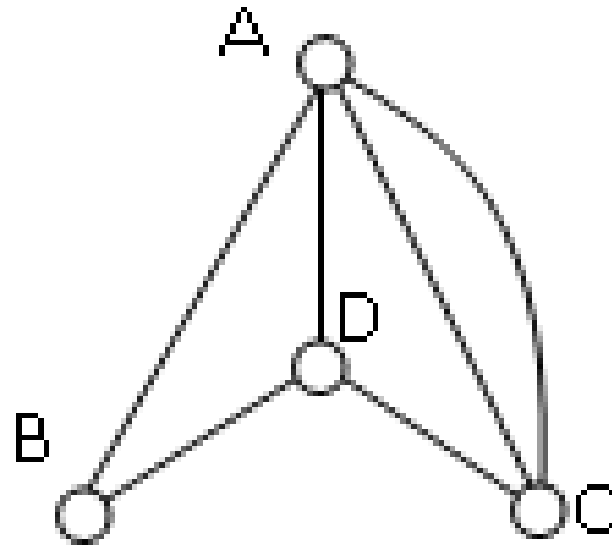
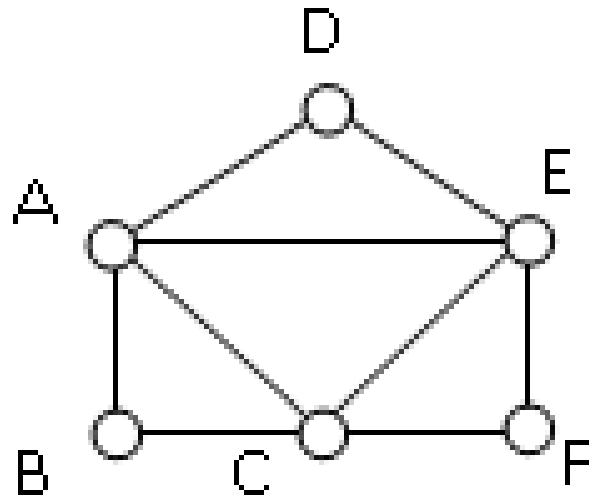


Euler Path, Euler Circuit or non-Traversable?

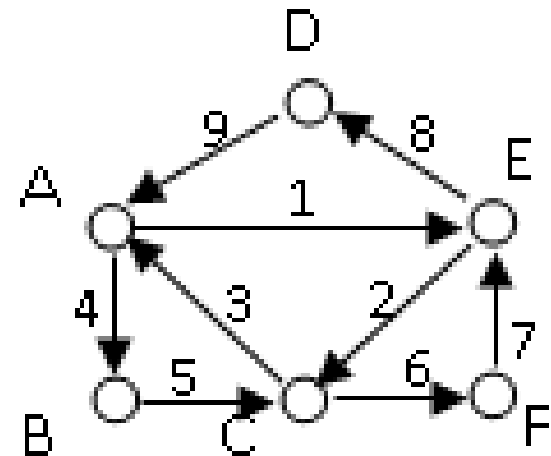


Euler Path, Euler Circuit or non-Traversable?



Now, find a circuit starting at A.

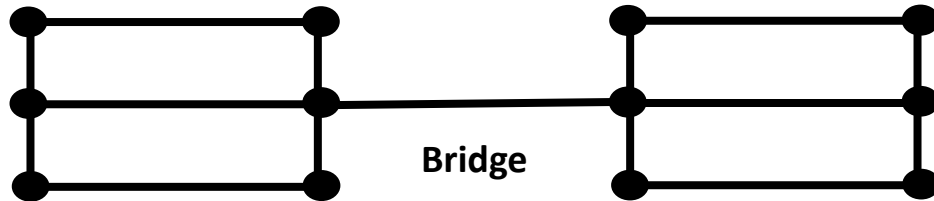
ADEACEFCBA and AECABCFEDA are two examples.



How do we find an Euler Path/Circuit, once we know it must exist?

- In a small graph, easy. In a more complicated graph, we have an **algorithm** to follow...a set of directions, like a map.
- This can become pretty complicated, but we will look at some easier graphs.

A bridge is the only edge connecting two separate sections of a graph.



Like with two odd vertices, we start at one end of the bridge, do our tracing, and then cross the bridge and finish tracing.

This concept of “not burning your bridges” is the idea behind the algorithm we will use for Euler Paths and Euler Circuits: **Fleury’s Algorithm**.

Fleury's Algorithm, formalized

- Start at any vertex if finding an Euler circuit. If finding an Euler path, start at one of the two vertices with odd degree.
- Choose any edge leaving your current vertex, making sure you aren't burning a bridge, or disconnecting the graph.
- Continue until you're done.

Fleury's Algorithm (Euler Circuit Only)

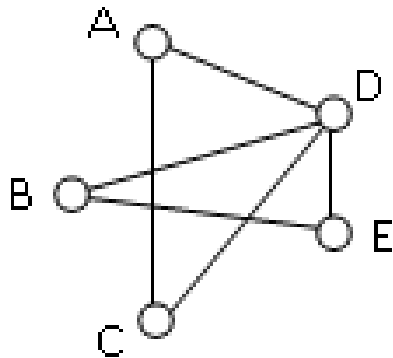
Ensure that **all vertices** have an even degree.

Ensure the graph is **connected** (you can reach any vertex from any other vertex)

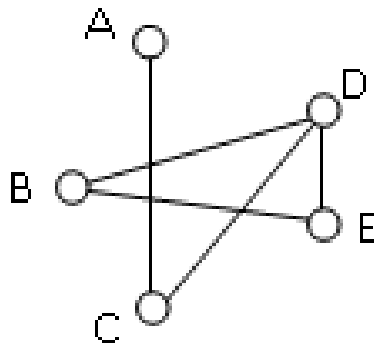
- **Start at Any Vertex:** Choose any vertex as the starting point, since all vertices have an even degree.
- **Traverse the Graph:** From the current vertex, move to any adjacent vertex through an edge. Choose an edge that is **not a bridge** unless it's the only remaining option.
- **Remove the Edge:** After traversing an edge, remove it from the graph so it isn't revisited.
- **Repeat Until All Edges Are Traversed:** Continue traversing edges and removing them, making sure to only cross bridges when no other edges are available.
- **Return to the Starting Vertex:** Since the graph contains an Euler Circuit, you will eventually return to the starting vertex after traversing all edges exactly once.

Use Fleury's algorithm to find an Euler Circuit, starting at vertex A.

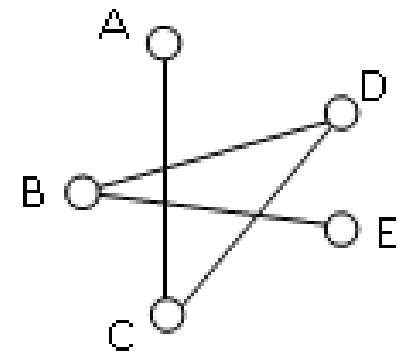
Original graph. We will choose edge AD



Next, from D we can choose to visit edge DB, DC or DE. But choosing edge DC will disconnect the graph (it is a bridge.) so we will choose DE.



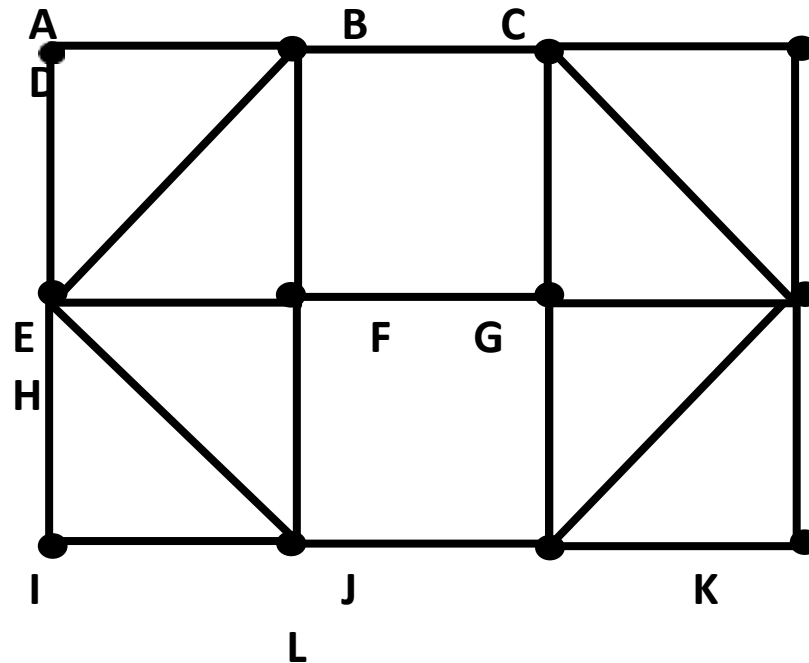
From vertex E, there is only one option and the rest of the circuit is determined.



Circuit: ADEBDCA

A,D,E,B,D,C,A

Fleury's algorithm: Euler path or circuit? Where to start?



Fleury's Algorithm (Euler path)

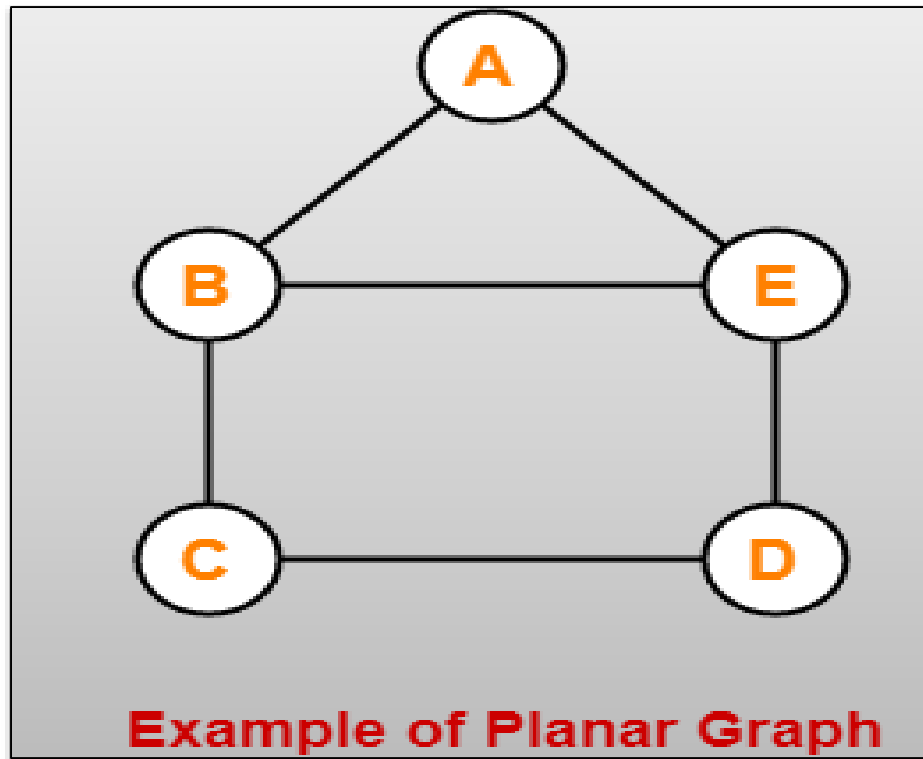
Will start from a vertex with odd degree.

All the remaining processes are similar.

Planar Graph-

Planar graph is a graph that can be drawn in a plane such that none of its edges cross each other.

The following graph is an example of a planar graph



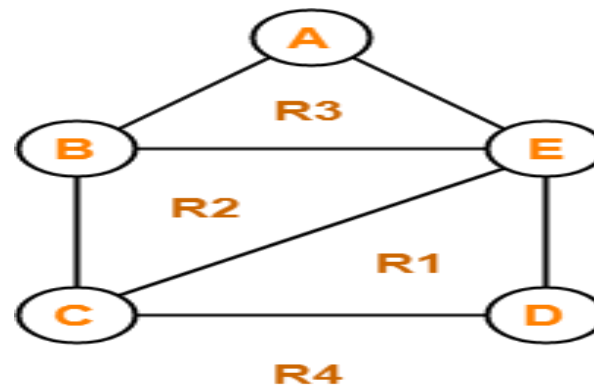
Here,
In this graph, no two edges cross each other.
Therefore, it is a planar graph.

Regions of Plane

- The planar representation of the graph splits the plane into connected areas called as **Regions of the plane**.
- Each region has some degree associated with it given as-
- Degree of Interior region = Number of edges enclosing that region
- Degree of Exterior region = Number of edges exposed to that region

Example

Consider the following planar graph



Regions of Plane

Here, this planar graph splits the plane into 4 regions- R1, R2, R3 and R4 where-

Degree (R1) = 3

Degree (R2) = 3

Degree (R3) = 3

Degree (R4) = 5

Planar Graph Properties

Property-01

In any planar graph, Sum of degrees of all the vertices = 2 x Total number of edges in the graph

$$\sum_{i=1}^n \deg(v_i) = 2 |E|$$

- **Property-02:** In any planar graph, Sum of degrees of all the regions = 2 x Total number of edges in the graph

$$\sum_{i=1}^n \deg(r_i) = 2 |E|$$

Theorem : *Euler's planar graph theorem*

For a **connected** planar graph :

$$v - e + r = 2$$

number
of vertices

number
of edges

number
of regions

- Property-03:
- If G is a planar graph with k components, then-
$$r + v - e = (k + 1)$$

- Problem-01:

Let G be a connected planar simple graph with 25 vertices and 60 edges. Find the number of regions in G .

- **Solution-**

Given-

Number of vertices (v) = 25

Number of edges (e) = 60

By Euler's formula, we know $r = e - v + 2$.

Substituting the values, we get-

Number of regions (r)

$$= 60 - 25 + 2$$

$$= 37$$

- Thus, Total number of regions in $G = 37$.

Problem-02:

Let G be a planar graph with 10 vertices, 3 components and 9 edges. Find the number of regions in G .

• **Solution-**

Given-

Number of vertices (v) = 10

Number of edges (e) = 9

Number of components (k) = 3

By Euler's formula, we know $r = e - v + (k+1)$

Substituting the values, we get-

Number of regions (r) = $9 - 10 + (3+1)$

= $-1 + 4$

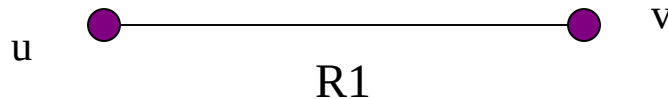
= 3

Thus, Total number of regions in G = 3.

Planar Graphs

- Proof of Euler's formula: By Induction

Base Case: for G_1 , $e_1 = 1$, $v_1 = 2$ and $r_1 = 1$

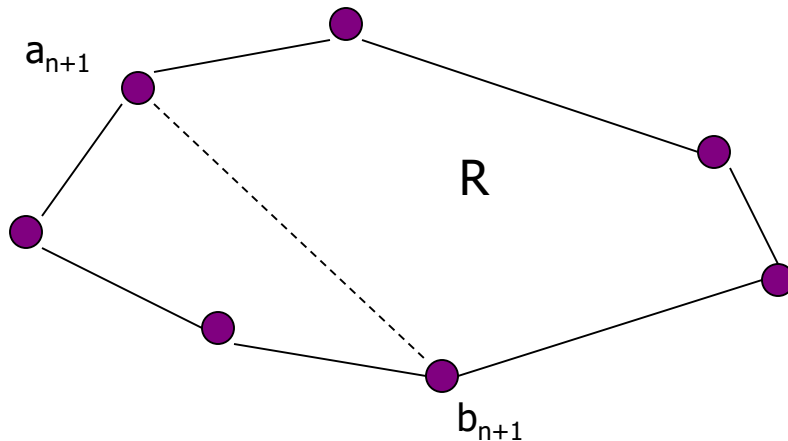


$n+1$ Case: Assume, $r_n = e_n - v_n + 2$ is true. Let $\{a_{n+1}, b_{n+1}\}$ be the edge that is added to G_n to obtain G_{n+1} and we prove that $r_n = e_n - v_n + 2$ is true. Can be proved using two cases.

Planar Graphs

- Case 1:

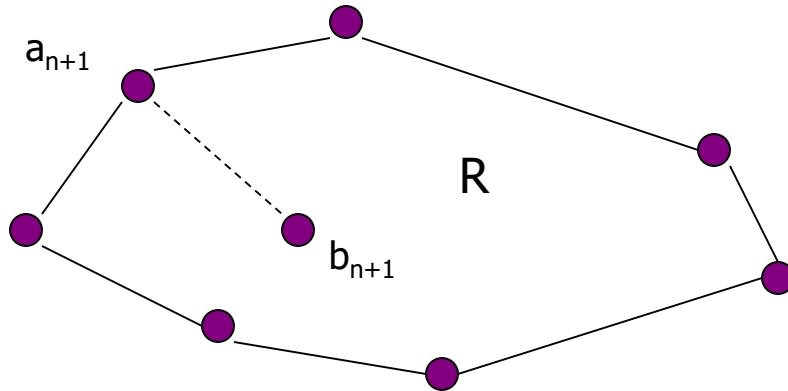
$$r_{n+1} = r_n + 1, e_{n+1} = e_n + 1, v_{n+1} = v_n \Rightarrow r_{n+1} = e_{n+1} - v_{n+1} + 2$$



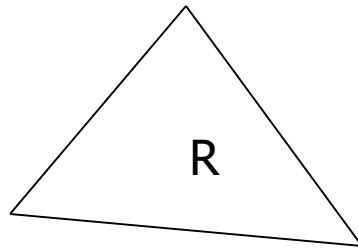
Planar Graphs

- Case 2:

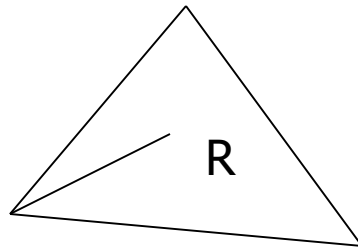
$$r_{n+1} = r_n, e_{n+1} = e_n + 1, v_{n+1} = v_n + 1 \Rightarrow r_{n+1} = e_{n+1} - v_{n+1} + 2$$



Region Degree



Degree of $R = 3$



Degree of $R = ?$

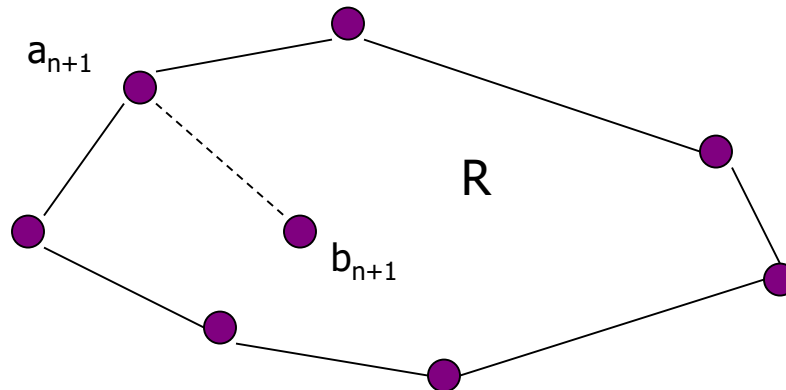
Planar Graphs

Corollary 1: Let $G = (V, E)$ be a connected simple planar graph with $|V| = v$, $|E| = e > 2$, and r regions. Then $3r \leq 2e$ and $e \leq 3v - 6$

Proof: Since G is loop-free and is not a multigraph, the boundary of each region (including the infinite region) contains at least three edges. Hence, each region has degree ≥ 3 .

Degree of region: No. of edges on its boundary; 1 edge may occur twice on boundary $\rightarrow$ contributes 2 to the region degree.

Each edge occurs exactly twice: either in the same region or in 2 different regions



Planar Graphs

Each edge occurs exactly twice: either in the same region or in 2 different regions

$\Rightarrow 2e = \text{sum of degree of } r \text{ regions determined by } 2e$

$\Rightarrow 2e \geq 3r$. (since each region has a degree of at least 3)

$\Rightarrow r \leq (2/3) e$

$\Rightarrow$ From Euler's theorem, $2 = v - e + r$

$\Rightarrow 2 \leq v - e + 2e/3$

$\Rightarrow 2 \leq v - e/3$

$\Rightarrow$ So $6 \leq 3v - e$

$\Rightarrow$ or $e \leq 3v - 6$

Planar Graphs

Corollary 2: Let $G = (V, E)$ be a connected simple planar graph then G has a vertex degree that does not exceed 5

Proof: If G has one or two vertices the result is true

If G has 3 or more vertices then by Corollary 1, $e \leq 3v - 6$

$$\Rightarrow 2e \leq 6v - 12$$

If the degree of every vertex were at least 6:

by Handshaking theorem: $2e = \text{Sum}(\text{deg}(v))$

$$\Rightarrow 2e \geq 6v. \text{ But this contradicts the inequality } 2e \leq 6v - 12$$

$\Rightarrow$ There must be at least one vertex with degree no greater than 5

Planar Graphs

Corollary 3: Let $G = (V, E)$ be a connected simple planar graph with v vertices ($v \geq 3$), e edges, and no circuits of length 3 then $e \leq 2v - 4$

Proof: Similar to Corollary 1 except the fact that no circuits of length 3 imply that degree of region must be at least 4.